



SAVE THE BEES

Interactive Data Visualization On Honey Bee Colony Loss

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INTRODUCTION



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Why we picked this topic?

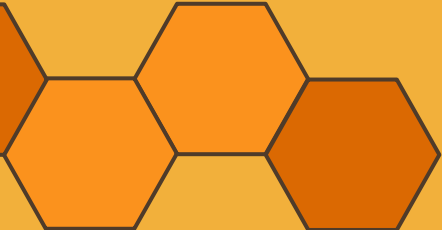
Honey bees are essential pollinators but face threats like habitat loss, pesticides, and diseases.

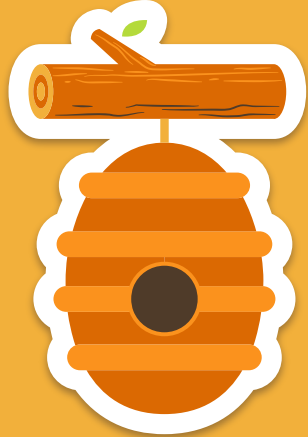
Objective?

Develop an interactive dashboard to analyze colony losses and identify patterns.

Fun Fact (not sure if this is a fact)

Humans would only survive for four years after all honey bees go extinct.





Dataset & Motivation !



Data Sources:

Save the Honey Bees dataset (colony loss data).

- US State & Territory Latitude & Longitude dataset (for mapping).

Why This Data?

- Offers insights into bee population decline across different states and years.
- Helps inform conservation policies.





Data Cleaning & Processing



01

**Merge Bee & US
GeoCode data**

02



**Removed
incomplete
entries**

03



**Stored structured data
in SQL/SQLite for easy
retrieval**



Live Demo Transition



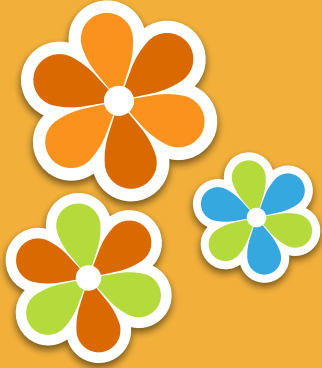
- What are the top causes of bee colony loss over time?
- Which states have the highest bee colony losses?
- What are the seasonal trends of colony loss?
- Is requeening effective in reducing loss?
- Which environmental factors correlate with colony loss?
- How does pesticide use impact colony loss?





01

**ITS DEMO
TIME!**





Dashboard Design

Overview

- **Layout:** Filters on top, visualizations below.
- **Color Palette:** Honeycomb (#FFC107), Bee Wings (#F5F5F5), Pollination Green (#4CAF50).
- **User Interaction:** Filters, hover-over details, sortable tables.
- **Tools Used:** Python (*Dash*, *Flask*), JavaScript (*D3.js*, *Plotly.js*)

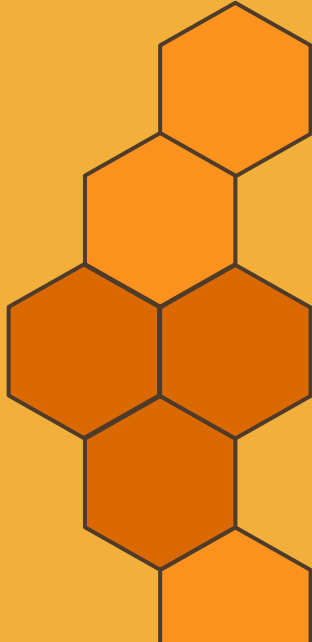


Research Questions Answered by the Dashboard

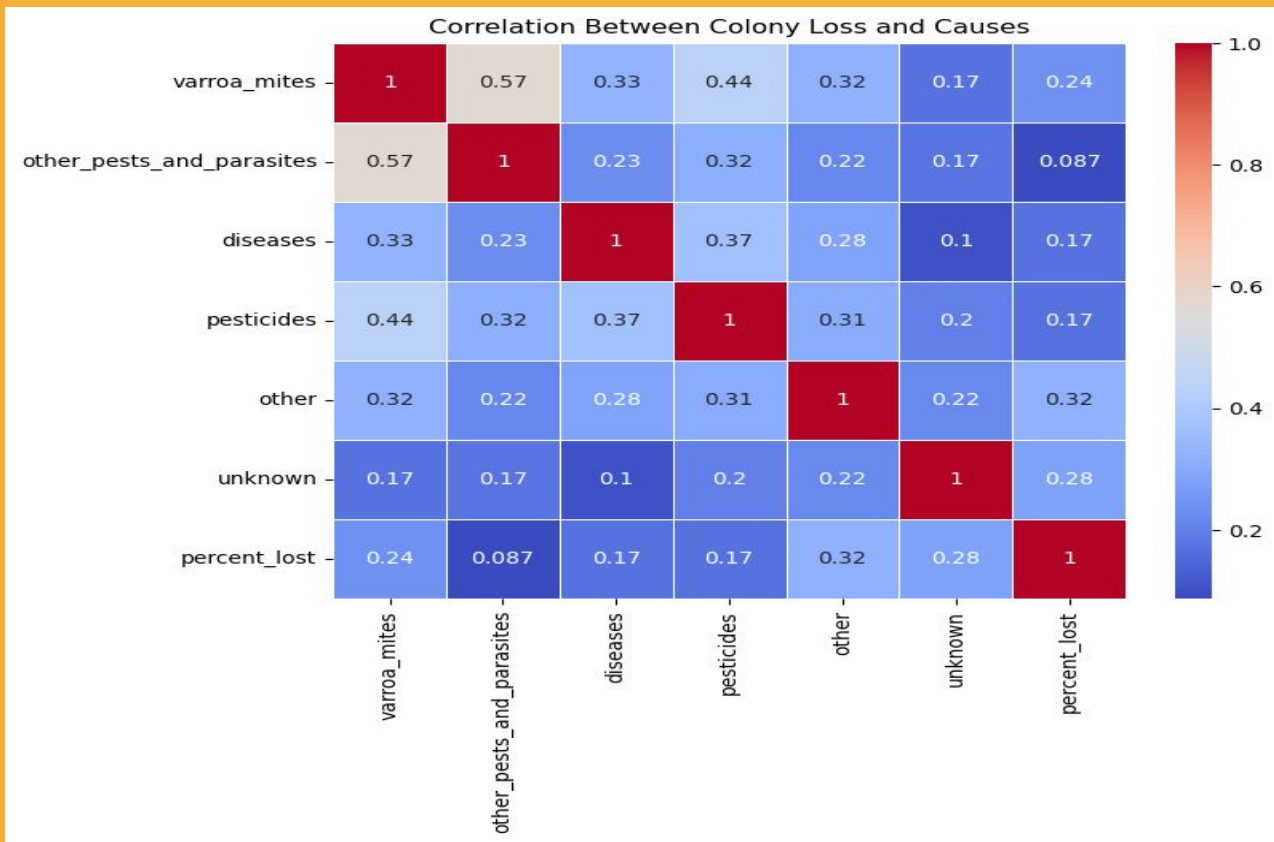


VISUAL INSIGHTS

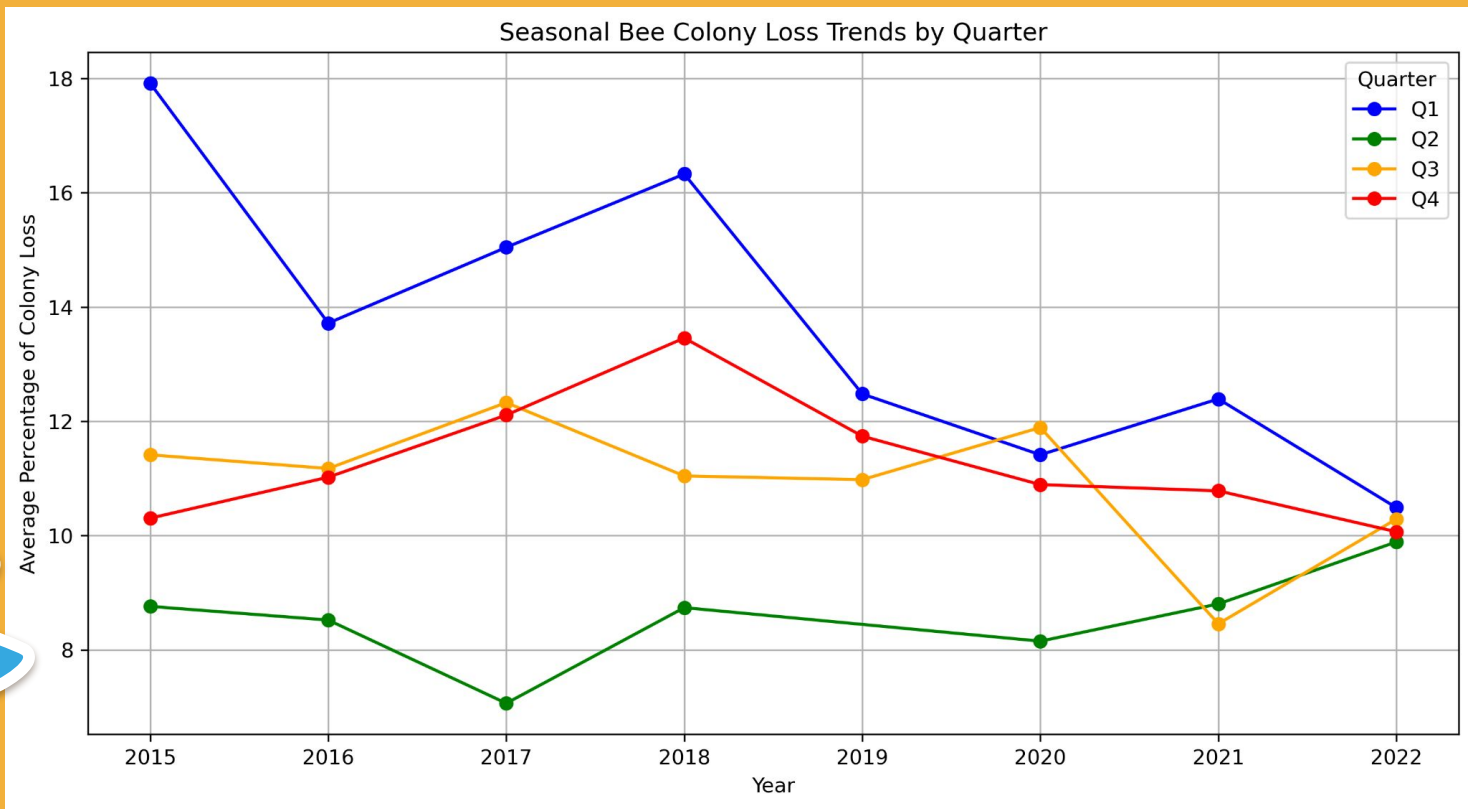
- Heatmap of colony loss over time.
- Choropleth map for health conditions per state.
- Scatter plot correlating pesticides & environmental factors to colony decline.



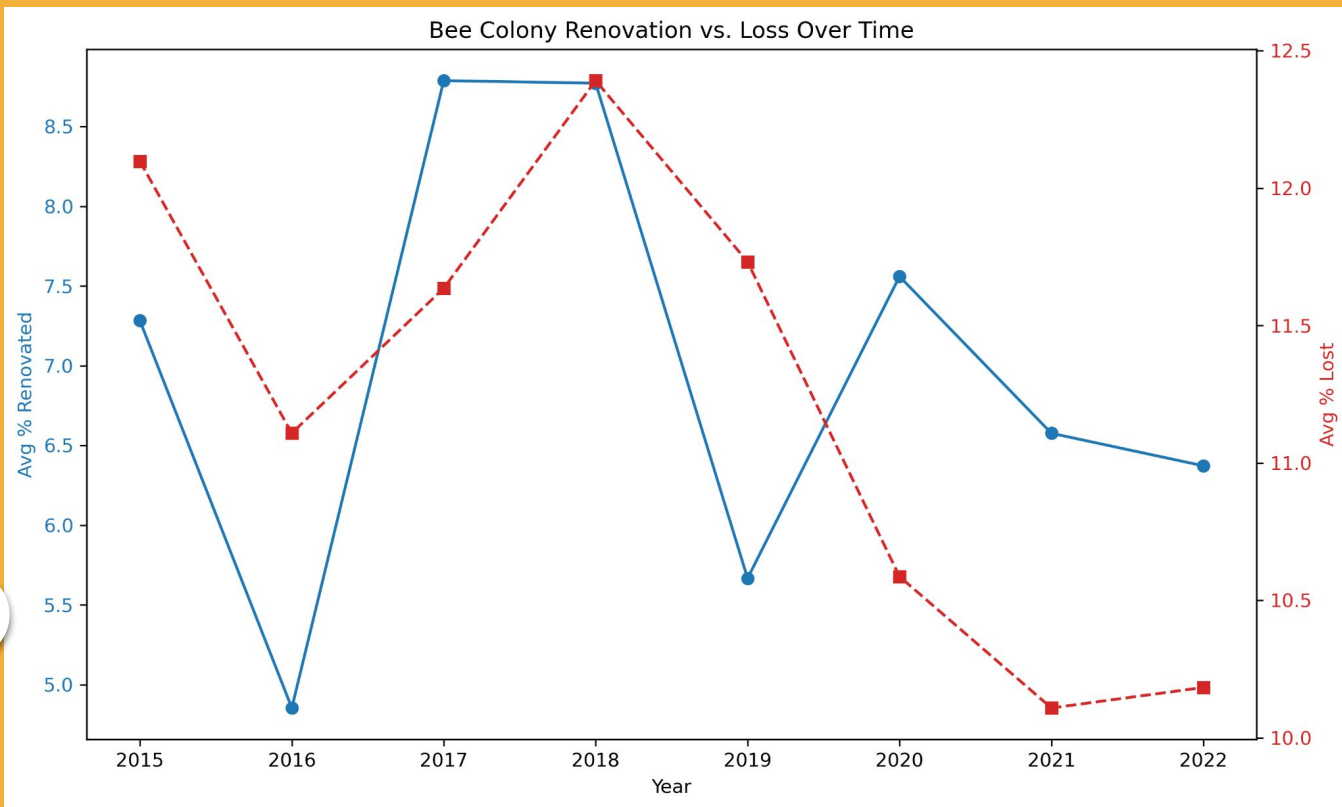
Correlation Between Colony Loss and Causes



What are the seasonal trends of colony loss?



Is requeening effective in reducing loss?





What impact does pesticide use have on bee colony loss?

1. Direct Toxicity

- **Immediate Death:** Bees can be directly poisoned by pesticides, especially if they come into contact with treated plants or if they forage on pesticide-contaminated flowers.

2. Behavioral Disruptions

- **Foraging and Navigation Problems:** Pesticides can disrupt bees' ability to forage and return to their hive. This is particularly problematic for honeybees, as they rely on a precise system of communication (the "waggle dance") to find food sources.

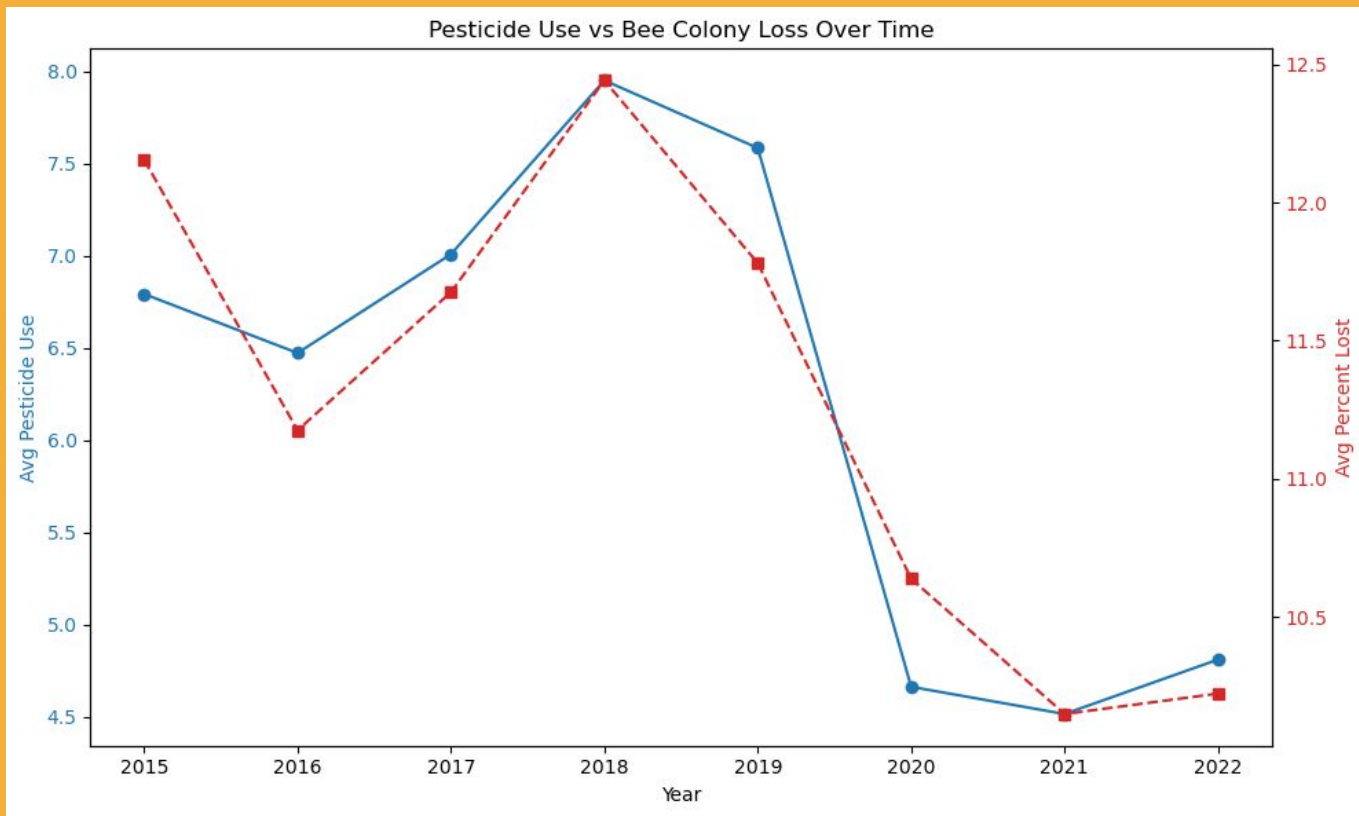
3. Immunosuppression

- **Vulnerability to Pathogens:** Pesticides can weaken bees' immune systems, making them more susceptible to diseases and parasites like the Varroa mite.

4. Loss of Worker Bees

- Worker bees are the backbone of the colony, responsible for gathering food and maintaining the hive. A decrease in their population due to pesticide exposure means the hive may not have enough workers to care for the queen and brood (larvae), leading to colony weakening and potential collapse.
- 

Pesticide Use vs Bee Colony Loss Over Time





Data Limitations & Bias

Challenges

- **Data Gaps:** Some states or years may have incomplete data, which could introduce gaps in the analysis. Additionally, some years might not be representative due to missing data.
- **Bias in Data Collection:** The data might be biased if certain states have more detailed reporting than others, or if certain factors (e.g., pesticide use or pest management) are underreported.
- **Sample Size:** The data might only represent a subset of the entire U.S., and results may not be generalizable to other regions or countries.





Conclusions & Future Work

Findings

- Significant losses in specific regions.
- Pesticides and diseases play a major role in colony decline.

Future Enhancements

- Integrate real-time data.
- Add advanced filtering & trend comparisons.
- Expand external data sources (temperature, land use, rainfall)





Call to Action

Encouraging Further Research:

- Support conservation efforts based on data-driven insights.
- Advocate for policy changes to protect pollinators.





Q&A

That concludes our presentation!
We'd love to hear any questions
you have.





AWESOME HONEY BEE

